

Godot Technical Blueprint

Architecture, scene tree, scripts, and JSON data examples

Architecture

The project should behave like a 2D arcade shooter but render as a 3D top-down scene. This gives modern visuals without simulator complexity.

Main scene tree

```
Main
├── WorldRoot
│   ├── Ocean
│   ├── IslandSpawner
│   ├── CloudLayer
│   ├── ShipLayer
│   ├── Player
│   ├── Enemies
│   ├── PlayerBullets
│   ├── EnemyBullets
│   └── Effects
├── EnemySpawner
├── StageManager
├── GameState
├── AudioManager
├── Camera3D
├── DirectionalLight3D
├── WorldEnvironment
└── HUD
```

Core script responsibilities

Script	Responsibility
main.gd	scene initialization, root references, high-level coordination
game_state.gd	score, lives, bombs, stage state, game over/mission complete
player.gd	arcade movement, bounds, damage state, signals
player_weapon.gd	fire cooldowns, weapon levels, bullet spawning
bullet_base.gd	projectile movement, damage, cleanup, hit handling
enemy_base.gd	health, score value, movement pattern, damage/death
enemy_spawner.gd	JSON wave loading and timed spawning
movement_patterns.gd	straight, diagonal, sine, boss intro, dive patterns
hud.gd	score/lives/bombs/stage/boss health display
audio_manager.gd	sound hooks and graceful missing-asset handling

Gameplay Principles

- Movement must be responsive and arcade-like. Do not implement realistic lift, drag, stalls, fuel, or long aircraft turn radius.
- The visual model can be large, but the damage hitbox should be small and fair.
- Bullets must remain readable even with realistic graphics. Use tracer glow, contrast, and consistent colour language.

- Stages should be data-driven. Do not hard-code entire levels into scripts.

Data Examples

enemy_types.json

```
{
  "fighter": {
    "scene": "res://scenes/enemies/EnemyFighter.tscn",
    "health": 2,
    "score_value": 100,
    "speed": 6.0,
    "fire_interval": 2.0,
    "drop_chance": 0.05
  },
  "heavy_fighter": {
    "scene": "res://scenes/enemies/EnemyFighter.tscn",
    "health": 5,
    "score_value": 250,
    "speed": 4.0,
    "fire_interval": 1.5,
    "drop_chance": 0.08
  },
  "bomber": {
    "scene": "res://scenes/enemies/EnemyBomber.tscn",
    "health": 12,
    "score_value": 800,
    "speed": 2.5,
    "fire_interval": 1.2,
    "drop_chance": 0.25
  },
  "boss_bomber": {
    "scene": "res://scenes/enemies/BossBomber.tscn",
    "health": 150,
    "score_value": 5000,
    "speed": 1.2,
    "fire_interval": 0.9,
    "drop_chance": 0.0
  }
}
```

stage_01_waves.json

```
[
  {
    "time": 1.0,
    "enemy_type": "fighter",
    "spawn_x": -4.0,
    "spawn_z": -18.0,
    "movement": "diagonal_right"
  },
  {
    "time": 1.3,
    "enemy_type": "fighter",
    "spawn_x": 4.0,
    "spawn_z": -18.0,
    "movement": "diagonal_left"
  },
  {
    "time": 3.0,
    "enemy_type": "fighter",
    "spawn_x": 0.0,
    "spawn_z": -18.0,
    "movement": "straight_down"
  },
  {
    "time": 5.0,
    "enemy_type": "fighter",
    "spawn_x": -2.5,
    "spawn_z": -18.0,
    "movement": "sine_wave"
  },
  {
    "time": 5.2,
    "enemy_type": "fighter",
    "spawn_x": 2.5,
```

```

    "spawn_z": -18.0,
    "movement": "sine_wave"
  },
  {
    "time": 8.0,
    "enemy_type": "bomber",
    "spawn_x": 0.0,
    "spawn_z": -20.0,
    "movement": "slow_center"
  },
  {
    "time": 13.0,
    "enemy_type": "heavy_fighter",
    "spawn_x": -5.0,
    "spawn_z": -18.0,
    "movement": "diagonal_right"
  },
  {
    "time": 13.0,
    "enemy_type": "heavy_fighter",
    "spawn_x": 5.0,
    "spawn_z": -18.0,
    "movement": "diagonal_left"
  },
  {
    "time": 20.0,
    "enemy_type": "boss_bomber",
    "spawn_x": 0.0,
    "spawn_z": -24.0,
    "movement": "boss_intro"
  }
]

```

weapon_types.json

```

{
  "level_1": {
    "name": "Dual machine guns",
    "bullet_count": 2,
    "fire_cooldown": 0.18,
    "spread_degrees": 0,
    "damage": 1
  },
  "level_2": {
    "name": "Fast dual guns",
    "bullet_count": 2,
    "fire_cooldown": 0.13,
    "spread_degrees": 4,
    "damage": 1
  },
  "level_3": {
    "name": "Triple shot",
    "bullet_count": 3,
    "fire_cooldown": 0.15,
    "spread_degrees": 12,
    "damage": 1
  },
  "level_4": {
    "name": "Wide spread shot",
    "bullet_count": 5,
    "fire_cooldown": 0.18,
    "spread_degrees": 22,
    "damage": 1
  },
  "rocket": {
    "name": "Rocket attack",
    "bullet_count": 1,
    "fire_cooldown": 0.8,
    "spread_degrees": 0,
    "damage": 8
  }
}

```

powerup_types.json

```

{
  "weapon_upgrade": {
    "label": "Weapon Upgrade",
    "effect": "increase_weapon_level",

```

```

    "duration": 0
  },
  "repair": {
    "label": "Repair",
    "effect": "restore_life_or_health",
    "duration": 0
  },
  "bomb": {
    "label": "Bomb",
    "effect": "add_bomb",
    "duration": 0
  },
  "score_bonus": {
    "label": "Score Bonus",
    "effect": "add_score",
    "amount": 1000,
    "duration": 0
  },
  "rapid_fire_temporary": {
    "label": "Rapid Fire",
    "effect": "temporary_fire_rate_boost",
    "duration": 8
  }
}

```

stage_01.json

```

{
  "id": "stage_01",
  "name": "Open Pacific",
  "wave_file": "res://data/waves/stage_01_waves.json",
  "environment": "open_pacific_day",
  "music": "stage_01_music",
  "boss": "boss_bomber",
  "target_duration_seconds": 180
}

```